

Final Project Report: Customer Churn Prediction

1. Introduction

Customer churn, the phenomenon of customers discontinuing their service with a company, represents a significant challenge for businesses, particularly in competitive sectors like telecommunications. This report details an end-to-end machine learning project aimed at predicting customer churn for a fictional telecommunications company. The primary objectives are to build predictive models, identify key drivers of churn, and propose actionable strategies for customer retention.

2. Data Overview and Preprocessing

The project utilizes the Telco Customer Churn dataset, originally sourced from IBM. The dataset comprises 7043 customer records, each with 20 features describing various aspects of customer demographics, services subscribed, and account information, along with a `Churn` target variable.

Data Cleaning Steps:

- TotalCharges Conversion:** The `TotalCharges` column, initially stored as an object type, was converted to a numeric format. Empty strings, representing new customers with no charges yet, were coerced to `NaN`.
- Missing Value Handling:** Rows with `NaN` values in `TotalCharges` (corresponding to customers with 0 tenure) were removed. This resulted in a cleaned dataset of 7032 records.
- SeniorCitizen Standardization:** The `SeniorCitizen` column, originally numerical (0 or 1), was mapped to categorical 'No' or 'Yes' for consistency with other binary categorical features.

The cleaned dataset, `customer_churn_cleaned.csv`, serves as the foundation for all subsequent analysis.

3. Exploratory Data Analysis (EDA)

EDA was conducted to uncover patterns and relationships within the data, focusing on the `Churn` variable. Key observations include:

- **Churn Distribution:** The dataset exhibits an imbalance, with approximately 26.5% of customers having churned.
- **Contract Type:** Customers on month-to-month contracts show a significantly higher churn rate compared to those with one-year or two-year contracts, indicating a lack of long-term commitment.
- **Tenure:** Shorter-tenured customers (0-12 months) have a much higher propensity to churn. Churn rates generally decrease as tenure increases.
- **Internet Service:** Customers with Fiber Optic internet service have a higher churn rate than those with DSL or no internet service.
- **Payment Method:** Electronic check users demonstrate the highest churn rates among all payment methods.
- **Security Services:** Customers without online security, online backup, device protection, and tech support services are more likely to churn, suggesting these services contribute to customer stickiness.
- **Monthly Charges:** While not a direct linear relationship, customers with higher monthly charges, especially those without bundled services, tend to churn more.

4. Feature Engineering

To prepare the data for machine learning models, the following feature engineering steps were performed:

- **customerID Removal:** The `customerID` column was dropped as it is a unique identifier and holds no predictive power.
- **Categorical Feature Encoding:** All categorical features were converted into numerical representations using **One-Hot Encoding**. The `drop='first'` strategy was applied to avoid multicollinearity.
- **Numerical Feature Scaling:** Numerical features (`tenure` , `MonthlyCharges` , `TotalCharges`) were scaled using `StandardScaler` to normalize their ranges, preventing features with larger values from dominating the model training.

5. Model Building and Evaluation

The dataset was split into an 80% training set and a 20% testing set, with stratification on the `Churn` variable to maintain the original class distribution. Three classification models were trained and evaluated:

- **Logistic Regression**

- **Decision Tree Classifier**
- **Random Forest Classifier**

Each model was evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC score. The ROC-AUC score was prioritized for model selection due to the imbalanced nature of the churn dataset.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.8045	0.6596	0.5511	0.6004	0.8441
Decision Tree	0.7306	0.4912	0.5057	0.4984	0.6780
Random Forest	0.7967	0.6403	0.4943	0.5579	0.8317

Note: The exact values may vary slightly based on random state and library versions.

Model Selection: Based on the ROC-AUC score, **Logistic Regression** emerged as the best-performing model in this specific run, demonstrating a good balance between identifying churners and non-churners. While Random Forest is generally powerful, Logistic Regression’s performance here suggests its linear decision boundary is effective for this dataset, and it offers better interpretability.

6. Feature Importance Analysis

For the best-performing model (Logistic Regression), coefficients can indicate feature importance. For ensemble models like Random Forest, `feature_importances_` attribute provides insights. Analyzing the feature importance (or coefficients for Logistic Regression) revealed that `Contract_Month-to-month`, `tenure`, `InternetService_Fiber optic`, `PaymentMethod_Electronic check`, `OnlineSecurity_No`, and `TechSupport_No` were among the most influential factors in predicting churn.

7. Business Insights and Recommendations

Based on the comprehensive analysis, the following business insights and recommendations are provided to help the telecommunications company reduce customer churn:

Business Insights:

- **Early Churn Risk:** New customers (low tenure) are highly susceptible to churn.

- **Contractual Commitment:** Month-to-month contracts are strongly associated with higher churn rates.
- **Payment Method Risk:** Customers using electronic checks have a higher propensity to churn.
- **Value-Added Services:** Lack of security and technical support services increases churn risk.
- **Internet Service Type:** Fiber optic users show higher churn, possibly due to higher expectations or competitive offerings.

Recommendations to Reduce Churn:

1. **Enhanced Onboarding Programs:** Implement robust onboarding for new customers, offering personalized support and incentives to foster early loyalty.
2. **Promote Long-Term Contracts:** Actively encourage customers to switch from month-to-month to longer-term contracts through attractive discounts or bundled offers.
3. **Optimize Electronic Check Experience:** Investigate and improve the electronic check payment process, or offer incentives for switching to more stable payment methods.
4. **Bundle and Highlight Value-Added Services:** Proactively market and bundle security and tech support services to increase perceived value and customer stickiness.
5. **Targeted Retention Campaigns:** Utilize the churn prediction model to identify high-risk customers and launch personalized retention campaigns, including special offers or proactive problem-solving.
6. **Customer Feedback Loop:** Establish mechanisms to gather feedback from churning customers to continuously improve services and address pain points.

8. Conclusion

This project successfully developed a customer churn prediction system, providing valuable insights into churn drivers and actionable recommendations. The Logistic Regression model demonstrated strong predictive capabilities, which can be deployed to proactively manage customer retention efforts. By focusing on the identified key areas, the telecommunications company can significantly reduce churn and improve customer lifetime value.

9. References

[1] IBM. (n.d.). *Telco Customer Churn*. Retrieved from <https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco->

Customer-Churn.csv